



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022–23
PRE BOARD EXAMINATION

CLASS: X
SUBJECT: MATHEMATICS [SET A]

FULL MARKS: 80
TIME: $2\frac{1}{2}$ HOURS

Answers to this Paper must be written on a paper provided separately.

You will not be allowed to write during the first 15 minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B.

The intended marks for questions or parts of questions are given in brackets [].

This paper contains 7 printed pages.

SECTION A

*(Attempt **all** questions from this section)*

Question 1

Choose the correct answers to the questions from the given options. (Do not copy the question, Write the correct answer only.) [15]

(i) The CGST paid by a customer to the shopkeeper for an article which is priced at ₹25000 is ₹ 2250. The rate of GST charged is:

- (a) 5% (b) 12% (c) 18% (d) 28%**

(ii) Which of the following equations has 2 as one of the roots?

- (a) $2x^2 - 7x + 6 = 0$ (b) $x^2 + 3x - 12 = 0$ (c) $x^2 - 4x + 5 = 0$ (d) $3x^2 - 6x - 2 = 0$**

(iii) If $\begin{bmatrix} a+b & 2 \\ 5 & b \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 2 \end{bmatrix}$, then the value of 'a' is:

- (a) 6 (b) 5 (c) 4 (d) 3**

(iv) $(x + 1)$ is not a factor of which of the given polynomials?

(a) $x^3 + x^2 - x + 1$ (b) $x^3 + x^2 + x + 1$ (c) $2x^3 + 3x^2 - 1$ (d) $3x^3 + 3x^2 + x + 1$

(v) The sum of the first 'n' terms of an A.P. is $3n^2 + 5n$, then the common difference of this A.P. is:

(a) 5 (b) 6 (c) 7 (d) 8

(vi) The point P(3, 5) on reflection in the line $y = -1$ is mapped onto P', the coordinates of P' is:

a) (2, 1) b) (-3, 1) c) (1, 4) d) (3, -7)

(vii) In triangle ABC, $DE \parallel BC$. If $AD = x$, $DB = x - 2$, $AE = x + 2$ and $EC = x - 1$, then the value of x is:

(a) 4 (b) 8 (c) 16 (d) 32

(viii) The diameter of a sphere is 6 cm. It is melted and drawn into a wire of diameter 0.2 cm. The length of the wire in metres is:

(a) 12m (b) 18m (c) 36m (d) 66m

(ix) If x is a negative integer, the solution set of $\frac{2}{3} + \frac{1}{3}(x + 1) > 0$ is:

(a) $\{-2, -3\}$ (b) $\{-1, -2\}$ (c) $\{-3, -4\}$ (d) $\{-4, -5\}$

(x) The letters of the word SOCIETY are placed at random in a row. The probability of getting a consonant is:

(a) $\frac{1}{7}$ (b) $\frac{2}{7}$ (c) $\frac{3}{7}$ (d) $\frac{4}{7}$

(xi) If $(x + 5)$ is the mean proportional between $(x + 2)$ and $(x + 9)$ the value of x is:

(a) 5 (b) 6 (c) 7 (d) 8

(xii) The line segment joining A(-2, 3) and B(6, -5) is intersected by X-axis at the point P. The coordinates of P are:

(a) (1, 0) (b) (0, 1) (c) (2, 0) (d) (-2, 0)

(xiii) The distance between two parallel tangents of a circle is 18 cm, then the radius of the circle is:

- (a) 8 cm (b) 10 cm (c) 9 cm (d) 7.5 cm

(xiv) The value of the expression $\sin^6\theta + \cos^6\theta + 3 \sin^2\theta \cos^2\theta$ is :

- (a) 0 (b) 1 (c) 2 (d) 3

(xv) For the following distribution:

Class Interval	0 - 10	10 -20	20 -30	30 - 40	40 - 50
Frequency	20	30	24	40	17

The sum of lower limits of the modal class and the median class is:

- (a) 20 (b) 30 (c) 40 (d) 50

Question 2

(i) A recurring deposit account of ₹ 500 per month has a maturity value of ₹ 9570. If the rate of interest is 8% per annum and interest is calculated at the end of every month, find the total time for which the account was held. [4]

(ii) Using properties of proportion, solve for x.

$$\frac{(m+n)x - (a-b)}{(m-n)x - (a+b)} = \frac{(m+n)x + (a+c)}{(m-n)x + (a-c)} \quad [4]$$

(iii) Prove that:

$$\left(\frac{1-\tan\theta}{1-\cot\theta} \right)^2 = \tan^2\theta \quad [4]$$

Question 3

(i) A toy is in the shape of a right circular cylinder with a hemisphere on one end and a cone on the other. The height and radius of the cylindrical part are 13cm and 5cm respectively. The radii of the hemispherical and conical parts are the same as that of the cylindrical part. Calculate the surface area of the toy if height of the conical part is 12cm. [4]

(ii) (-2, -1) and (4, -5) are the coordinates of the vertices B and D respectively of a rhombus ABCD. Find the equation of the diagonal AC. [4]

- (iii) Use a graph paper for this question taking 1cm = 1unit along both the x- axis and y- axis. Plot the points A(3, 0) and B(0, 4).
 a) Write down the coordinates of A', the reflection of A in the y- axis.
 b) Write down the coordinates of B', the reflection of B in the x- axis.
 c) Assign a special name to quadrilateral ABA'B'A.
 d) If C is the midpoint of AB, write down the coordinates of C' the reflection of C in the origin. [5]

SECTION B

(Attempt *any four* questions from this section)

Question 4

- (i) Solve the following equation for x and give your answer correct to 3 significant figures:
 $x^2 - 3x = 9$ [3]

- (ii) Mr Sunil Sonare visits the market and buys the following articles:

A linen shirt costing ₹ 3000, GST @ 12%

A smart watch costing ₹ 2000 with a discount of 30%, GST @ 18%.

Calculate the total bill amount including GST paid by Mr Sonare. [3]

- (iii) The table below shows the daily expenditure on food of 50 house-holds in a locality.

Daily Expenditure(in ₹)	0 -100	100 -200	200 - 300	300 - 400	400 - 500	500 - 600
Number of House-holds	5	8	15	10	7	5

Using a graph paper, draw a histogram representing the above distribution and estimate the mode. Take along x-axis 2cm = ₹100 and along y- axis 2cm = 2 Households. [4]

Question 5

- (i) Find the equation of a line passing through the intersection of $2x - y = 1$ and $3x + 2y + 9 = 0$ and making an angle of 45° with the positive direction of the X axis. [3]

- (ii) Prove the following identity:

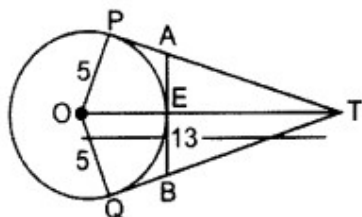
$$\cot^2 A \left(\frac{\sec A - 1}{1 + \sin A} \right) + \sec^2 A \left(\frac{\sin A - 1}{1 + \sec A} \right) = 0 \quad [3]$$

- (iii) The sum of the third and seventh terms of an A.P. is 6 and their product is 8. Find the sum of the first sixteen terms of the A.P. [4]

Question 6

(i) If HCF of $(x - 5)(x^2 - x - a)$ and $(x - 4)(x^2 - 2x - b)$ is $(x - 4)(x - 5)$, find the values of a and b . [3]

(ii) In the figure, O is the centre of a circle of radius 5 cm. T is a point such that $OT = 13$ cm and OT intersects circle at E . If AB is a tangent to the circle at E , find the length of AB , where TP and TQ are two tangents to the circle.



[3]

(iii) If $X = \begin{bmatrix} \sin 90^\circ & \sec 60^\circ \\ \operatorname{cosec} 30^\circ & \cos 0^\circ \end{bmatrix}$ and I is the unit matrix of the same order as X ,

then show that $X^2 - 2X = 3I$.

[4]

Question 7

(i) A game of cards numbered 1 to 40, one card is drawn at random. Find the probability that the card drawn has a number:

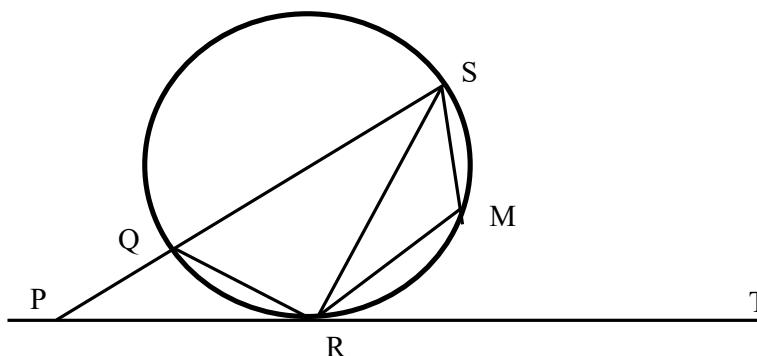
- a) a perfect square
- b) a prime number
- c) a multiple of 2 and 3

[3]

(ii) Wax cylinder of diameter 21cm and height 21cm is chipped off and shaped to form a cone of maximum volume. The chipped off wax is recast into a solid sphere. Find the diameter of the sphere. [3]

(iii) In the given figure, QS is a diameter of the circle. PT touches the circle at R . If, $\angle PRQ = 20^\circ$, $\angle SPR = 50^\circ$ and $\angle RSM = 35^\circ$, find:

- (a) $\angle SQR$
- (b) $\angle PSR$
- (c) $\angle SMR$
- (d) $\angle MRT$



[4]

Question 8

(i) Solve the following inequation and represent the solution set on the number line.

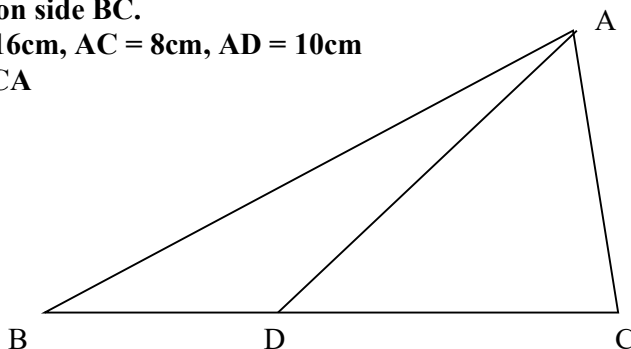
$$-\frac{1}{2} \leq \frac{9x}{10} + 4 \leq \frac{2}{5}, x \in \mathbb{R} \quad [3]$$

(ii) In $\triangle ABC$, D is a point on side BC.

$$\angle ABC = \angle DAC, AB = 16\text{cm}, AC = 8\text{cm}, AD = 10\text{cm}$$

a) Prove that $\triangle ACD \sim \triangle BCA$

b) Find BC and CD



[3]

(iii) A factory manufactures nuts and bolts of various sizes. The measurement of inner diameters of 1000 nuts is given in the following frequency table. Determine the mean inner diameter per nut using the short cut method.

Diameter(mm)	43 - 45	46 - 48	49 - 51	52 - 54	55 - 57
No: of nuts	175	236	200	196	193

[4]

Question 9

(i) A trader buys 'x' articles for a total cost of ₹600.

a) Write down the cost of one article in terms of 'x'. If the cost per article was ₹5 more, the number of articles that can be bought for ₹600 would be four less.

b) Write down the equation in 'x' for the above situation and solve it to find 'x'. [4]

(ii) Use graph paper to answer this question.

The table given below shows the salaries of 100 persons.

Salary(in thousand ₹)	5000 – 10,000	10000- 15000	15000- 20000	20000- 25000	25000- 30000	30000- 35000	35000- 40000	40000- 45000	45000- 50000
No. of persons	4	7	10	12	16	14	20	11	6

Draw an ogive of the given data.

Hence use your graph to find the following:

a) Median

b) Lower Quartile

c) Number of persons whose salary is above ₹28,000.

d) Persons earning between ₹18,000 and ₹24,000 are thinking of changing their jobs.

Find how many persons are planning for a new job. [6]

Question 10

(i) $\frac{x}{b-c} = \frac{y}{c-a} = \frac{z}{a-b}$, prove that

a) $x + y + z = 0$ b) $ax + by + cz = 0$ [3]

(ii) Using ruler and compasses construct a triangle ABC where $AB = 3\text{cm}$, $BC = 4\text{cm}$ and $\angle ABC = 90^\circ$. Hence construct a circle circumscribing the triangle ABC. Measure and write down the radius of the circle. [3]

(iii) The horizontal distance between two towers is 150m. The angle of elevation of the top and the angle of depression of the bottom of the first tower as observed from the top of the second tower is 30° and 26° respectively. Find the heights of the two towers. Give your answer correct to 3 significant figures. [4]